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Calibration method, detection system, exposure apparatus, article manufacturing method, and non-transitory computer-readable storage medium

Abstract

A calibration method of a detection system including an illumination system configured to illuminate a detection target, and an imaging system configured to form an image of light from the detection target on a photoelectric conversion element, the method including obtaining, for each of at least two combinations of first apertures in the illumination system and second apertures in the imaging system, each of which is formed by selecting one first aperture and one second aperture from the plurality of first apertures and the plurality of second apertures, a defocus characteristic indicating a shift amount of the image on the photoelectric conversion element with respect to a defocus amount of the detection target in a state in which each of the first aperture and the second aperture is positioned in a first position shifted from a reference position.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8049891	12/2010	Maeda	N/A	N/A
11231573	12/2021	Fujishima et al.	N/A	N/A
2004/0036879	12/2003	Fukui	356/401	G03F 9/7088
2012/0123581	12/2011	Smilde	706/12	G03F 7/70483
2015/0227061	12/2014	Tinnemans	356/509	G03F 9/7069
2019/0346771	12/2018	Pandey	N/A	G03F 7/70133
2020/0319447	12/2019	Fujishima	N/A	G02B 21/361

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
4944690	12/2011	JP	N/A
2020170070	12/2019	JP	N/A

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a calibration method, a detection system, an exposure apparatus, an article manufacturing method, and a non-transitory computer-readable storage medium.

Description of the Related Art

(2) In recent years, an exposure apparatus used for the manufacture of a semiconductor device or the like is demanded to improve overlay accuracy between an original and a substrate along with further miniaturization of the resolution. Since the overlay accuracy corresponding to about ⅕ of the resolution is usually required, improvement of the overlay accuracy becomes more and more important as the miniaturization of a semiconductor device advances.

(3) To improve the overlay accuracy, there is, for example, a method of accurately adjusting a position detection system of an exposure apparatus. Generally, the position detection system is optically configured, and includes an illumination system for illuminating a detection target (for example, a mark) formed on a substrate with detection (observation) light, and an imaging system for detecting an image of the detection target by concentrating light from the detection target. To implement a good imaging state of the position detection system, adjusting the position of an aperture stop, that is, a pupil formed in each of the illumination system and the imaging system is one of important technical problems requiring high positioning accuracy. Therefore, each of Japanese Patent No. 4944690 and Japanese Patent Laid-Open No. 2020-170070 has proposed a technique that adjusts the position of the aperture stop of at least one of the illumination system and the imaging system of the position detection system.

(4) Unfortunately, maintaining the pupil position adjusted by using, for example, the technique disclosed in Japanese Patent No. 4944690 or Japanese Patent Laid-Open No. 2020-170070 in the illumination system or the imaging system of the position detection system over an operation period of the exposure apparatus on the user site is also an important technical problem different from the initial adjustment. Generally, to cope with the limit of holding of the aperture stop resulting from the mechanical accuracy and the reset of the exposure apparatus including power shutdown, the pupil position adjusted in the illumination system or the imaging system of the detection system is stored by using an encoder. However, the use of the encoder is costly, and, in addition to that, the encoder itself generates heat, and this may fluctuate the refractive index of air in the exposure apparatus. Also, unlike the initial adjustment in the manufacturing process of the exposure apparatus, tools and the time for adjustment are largely restricted on the user site. Accordingly, in the illumination system and the imaging system of the position detection system, a technique capable of returning to the initially adjusted pupil position in a self-contained manner in the exposure apparatus without using an encoder is demanded.

SUMMARY OF THE INVENTION

(5) The present invention provides a technique advantageous in calibrating a detection system for detecting the position of a detection target.

(6) According to one aspect of the present invention, there is provided a calibration method of a detection system including an illumination system configured to illuminate a detection target, and an imaging system configured to form an image of light from the detection target on a photoelectric conversion element, wherein the illumination system includes a plurality of first apertures selectively arranged in a pupil plane of the illumination system and having different openings, the imaging system includes a plurality of second apertures selectively arranged in a pupil plane of the imaging system and having different openings, and the method includes obtaining, for each of at least two combinations of first apertures and second apertures, each of which is formed by selecting one first aperture and one second aperture from the plurality of first apertures and the plurality of second apertures, a first defocus characteristic indicating a shift amount of the image on the

photoelectric conversion element with respect to a defocus amount of the detection target in a state in which each of the first aperture and the second aperture is positioned in a first position shifted from a reference position, and performing position adjustment on each of the first aperture and the second aperture such that each of the first aperture and the second aperture is positioned in the reference position, based on a reference defocus characteristic indicating a shift amount of the image on the photoelectric conversion element with respect to a defocus amount of the detection target in a state in which each of the first aperture and the second aperture is positioned in the reference position, and on the first defocus characteristic obtained in the obtaining.
(7) Further aspects of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic view illustrating configurations of an exposure apparatus according to an aspect of the present invention.
(2) FIG. **2** is a view illustrating detailed configurations of a stage reference plate.
(3) FIG. **3** is a view illustrating detailed configurations of a position detection system.
(4) FIG. **4** is a view illustrating a practical configuration of a switching mechanism.
(5) FIG. **5** is a view illustrating a practical configuration of a switching mechanism.
(6) FIG. **6** is a view illustrating an example of the state of the position detection system.
(7) FIG. **7** is a view illustrating an example of the relationship between the driving amount of a substrate stage and the fluctuation amount of a defocus characteristic.
(8) FIG. **8** is a view illustrating an example of an alignment mark.
(9) FIG. **9** is a flowchart for explaining a process (calibration method) of calibrating the position detection system.
DESCRIPTION OF THE EMBODIMENTS
(10) Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed invention. Multiple features are described in the embodiments, but limitation is not made to an invention that requires all such features, and multiple such features may be combined as appropriate. Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.
(11) FIG. **1** is a schematic view illustrating configurations of an exposure apparatus **100** according to an aspect of the present invention. The exposure apparatus **100** is a lithography apparatus that exposes a substrate via an original to form a pattern on the substrate. The exposure apparatus **100** includes an original stage **2** that holds an original **1** (reticle or mask), a substrate stage **4** that holds a substrate **3**, and an illumination optical system **5** that illuminates the original **1** held by the original stage **2**. The exposure apparatus **100** also includes a projection optical system **6** that projects (the image of) the pattern of the original **1** to the substrate **3** held by the substrate stage **4**, and a control unit **17** that comprehensively controls the overall operation of the exposure apparatus **100**.
(12) In this embodiment, the exposure apparatus **100** is a scanning exposure apparatus (scanner) that transfers the pattern of the original **1** to the substrate **3** while synchronously scanning the original **1** and the substrate **3** in the scanning direction (that is, by a step & scan method). However, the exposure apparatus **100** may be an exposure apparatus (stepper) that transfers the pattern of the original **1** to the substrate **3** while fixing the original **1** (that is, by the step & scan method).
(13) In the following description, as illustrated in FIG. **1**, a direction (optical axis direction) coincident with the optical axis of the projection optical system **6** will be defined as the Z direction.

The scanning direction of the original **1** and the substrate **3** in a plane perpendicular to the Z direction will be defined as the Y direction. A direction (non-scanning direction) perpendicular to the Z and Y directions will be defined as the X direction. Directions around the X-, Y-, and Z-axes will be defined as OX, OY, and OZ directions, respectively.
(14) The illumination optical system **5** illuminates the original **1**, more specifically, a predetermined illumination region on the original with light (exposure light) of a uniform illuminance distribution. Examples of the exposure light are the g-ray and i-ray of ultra-high pressure mercury lamps, a KrF excimer laser, an ArF excimer laser, and an F.sub.2 laser. To manufacture a smaller semiconductor element, Extreme UltraViolet light (EUV light) of several nm to several hundred nm may be used as the exposure light.
(15) The original stage **2** is configured to be two-dimensionally movable in a plane perpendicular to the optical axis of the projection optical system **6**, that is, in the X-Y plane and be rotatable in the OZ direction while holding the original **1**. A driving mechanism (not shown) such as a linear motor drives the original stage **2**.
(16) A mirror **7** is provided on the original stage **2**. A laser interferometer **9** is provided at a position facing the mirror **7**. The laser interferometer **9** measures in real time the two-dimensional position and rotation angle of the original stage **2** (the original **1** held by the original stage **2**), and outputs the measurement result to the control unit **17**. The control unit **17** controls the driving mechanism based on the measurement result of the laser interferometer **9**, and positions the original **1** held by the original stage **2**.
(17) The projection optical system **6** includes a plurality of optical elements, and projects the pattern of the original **1** to the substrate **3** at a predetermined projection magnification β. In this embodiment, the projection optical system **6** is a reduction optical system having the projection magnification β of, for example, ¼ or ⅕.
(18) The substrate stage **4** includes a Z stage that holds the substrate **3** via a chuck, an X-Y stage that supports the Z stage, and a base that supports the X-Y stage. A driving mechanism **18** including a linear motor and the like drives the substrate stage **4**.
(19) A mirror **8** is provided on the substrate stage **4**. Laser interferometers **10** and **12** for measuring the position of the substrate stage **4** are provided at positions facing the mirror **8**. The laser interferometer **10** measures positions of the substrate stage **4** in the X direction, Y direction, and OZ direction in real time, and outputs the measurement result to the control unit **17**. The laser interferometer **12** measures positions of the substrate stage **4** in the Z direction, OX direction, and OY direction in real time, and outputs the measurement result to the control unit **17**. The control unit **17** controls the driving mechanism **18** based on the measurement results of the laser interferometers **10** and **12**, thereby positioning the substrate **3** held by the substrate stage **4**.
(20) A stage reference plate **11** is provided on the substrate stage **4** so as to be almost flush with the surface of the substrate **3** held by the substrate stage **4**. FIG. **2** is a view illustrating the detailed configurations of the stage reference plate **11** provided on the substrate stage **4**. The stage reference plate **11** may be provided at one corner of the substrate stage **4**, or may be provided at each of a plurality of corners of the substrate stage **4**. Alternatively, the stage reference plate **11** may be provided along a side of the substrate stage **4**. In this embodiment, as illustrated in FIG. **2**, three stage reference plates **11** are provided on the substrate stage **4**.
(21) As illustrated in FIG. **2**, the stage reference plate **11** includes a reference mark **111** to be detected by original position detection systems **13** and **14**, and a reference mark **112** to be detected by a position detection system **16**. The stage reference plate **11** may include a plurality of the reference marks **111** and a plurality of the reference marks **112**. The positional relationship (X and Y directions) between the reference mark **111** and the reference mark **112** is set to a predetermined positional relationship (that is, it is known). Note that the reference mark **111** and the reference mark **112** may be common marks.
(22) The original position detection system **13** is provided near the original stage **2**. The original

position detection system **13** detects an original reference mark (not shown) provided on the original **1** held by the original stage **2**, and the reference mark **111** provided on the stage reference plate **11** on the substrate stage through the projection optical system **6**. The original position detection system **13** detects the original reference mark provided on the original **1**, and the reference mark **111** through the projection optical system **6** by using the same light source as one used when actually exposing the substrate **3**. More specifically, the original position detection system **13** detects, by an image sensor (for example, a photoelectric conversion element such as a CCD camera), beams reflected by the original reference mark and the reference mark **111** (reflective mark). The original **1** and the substrate **3** are aligned based on a detection signal from the image sensor. At this time, the position and focus are adjusted between the original reference mark provided on the original **1** and the reference mark **111** provided on the stage reference plate **11**. As a result, the relative positional relationship (X, Y, and Z) between the original **1** and the substrate **3** can be adjusted.
(23) The original position detection system **14** is provided on the substrate stage **4**. The original position detection system **14** is a transmission detection system and is used when the reference mark **111** is a transmission mark. The original position detection system **14** detects the original reference mark provided on the original **1** and the reference mark **111** provided on the stage reference plate **11** by using the same light source as one used when actually exposing the substrate **3**. More specifically, the original position detection system **14** detects, by a light amount sensor, the transmitted light having passed through the original reference mark and the reference mark **111**. At this time, the original position detection system **14** detects the amount of transmitted light while moving the substrate stage **4** in the X direction (or Y direction) and the Z direction. Accordingly, the position and focus can be adjusted between the original reference mark provided on the original **1** and the reference mark **111** provided on the stage reference plate **11**.
(24) In this way, the original position detection system **13** or the original position detection system **14** can be arbitrarily used to adjust the relative positional relationship (X, Y, and Z) between the original **1** and the substrate **3**.
(25) A focus detection system **15** includes a light projecting system that obliquely projects light to the surface of the substrate **3**, and a light receiving system that receives light reflected by the surface of the substrate **3**. The focus detection system **15** detects the Z-direction (height-direction) position of the substrate **3**, and outputs the detection result to the control unit **17**. The control unit **17** controls the driving mechanism **18** based on the detection result of the focus detection system **15** to adjust the Z-direction position (focus position) and inclination angle of the substrate **3** held by the substrate stage **4**.
(26) The position detection system **16** includes an illumination system that illuminates an alignment mark **19** as a detection target on a substrate and the reference mark **112** of the stage reference plate **11**, and an imaging system that forms images of light (reflected light) from the alignment mark **19** and the reference mark **112** on a photoelectric conversion element. The position detection system **16** detects the alignment mark **19** and the reference mark **112**, and outputs the detection results to the control unit **17**. The control unit **17** controls the driving mechanism **18** based on the detection result (the position of the alignment mark **19**) of the position detection system **16** to drive the substrate stage **4**, thereby adjusting the X- and Y-direction positions of the substrate **3** held by the substrate stage **4**.
(27) In general, the arrangement of an optical position detection system for detecting (observing) an alignment mark on a substrate is roughly divided into two: an off-axis detection system and a Through The Lens (TTL) detection system. The off-axis detection system optically detects an alignment mark provided on a substrate without the intervention of a projection optical system. The TTL detection system detects an alignment mark provided on a substrate through a projection optical system by using light (non-exposure light) different in wavelength from exposure light. Although the position detection system **16** is the off-axis detection system in this embodiment, the

detection method is not limited. For example, when the position detection system **16** is the TTL detection system, it detects the alignment mark **19** provided on the substrate **3** through the projection optical system **6**. Except this, the basic arrangement is the same as that of the off-axis detection system.
(28) FIG. **3** is a view illustrating the detailed configurations of the position detection system **16** including an illumination system ILS and an imaging system IMS. The position detection system **16** includes a light source **20**, a first illumination optical system **21**, a wavelength filter plate **22**, a second illumination optical system **23**, an aperture stop plate **24**, a third illumination optical system **25**, a fourth illumination optical system **26**, a polarizing beam splitter **27**, an NA stop **28**, and a prism **29**. The position detection system **16** also includes a λ/4 plate **30**, an objective lens **31**, a relay lens **32**, a first imaging optical system **33**, an aperture stop plate **34**, a second imaging optical system **35**, a wavelength shift difference adjustment optical member **36**, a photoelectric conversion element **37**, and switching mechanisms **38** and **39**.
(29) In this embodiment, the light source **20** emits visible light (for example, light having a wavelength of 500 nm or more to 700 nm or less), light of the blue wavelength (for example, light having a wavelength of 450 nm or more to 550 nm or less (blue wavelength light)), and infrared light (for example, light having a wavelength of 700 nm or more to 1,500 nm or less). The light (illumination light) emitted by the light source **20** passes through the first illumination optical system **21**, the wavelength filter plate **22**, and the second illumination optical system **23**, and reaches the aperture stop plate **24** positioned on the pupil plane (optical Fourier transform plane with respect to the object plane) of the position detection system **16**.
(30) The wavelength filter plate **22** includes a plurality of wavelength filters different in the wavelength band (transmission wavelength band) of light to be transmitted. One wavelength filter is selected from the plurality of wavelength filters under the control of the control unit **17**, and is inserted in the optical path of the position detection system **16**. In this embodiment, the wavelength filter plate **22** includes a wavelength filter that transmits visible light, a wavelength filter that transmits blue wavelength light, and a wavelength filter that transmits infrared light. By switching between these wavelength filters in the wavelength filter plate **22**, the wavelength band of light for illuminating the alignment mark **19** provided on the substrate **3** can be selected. The wavelength filter plate **22** may be configured so that a new wavelength filter can be added in addition to the plurality of wavelength filters provided in advance.
(31) The aperture stop plate **24** includes a plurality of aperture stops **24*a*** (first apertures) having different numerical apertures (aperture (opening) sizes, NA) and different aperture (opening) shapes. The plurality of aperture stops **24*a*** are selectively arranged in the pupil plane of the illumination system ILS. In this embodiment, an illumination mode for illuminating the alignment mark **19** can be changed by switching the optical paths of the illumination system ILS of the position detection system **16**, more specifically, by switching the aperture stops **24*a*** to be arranged in the pupil (pupil plane), under the control of the control unit **17**. The aperture stop plate **24** can also have a configuration capable of adding a new aperture stop, in addition to the plurality of aperture stops **24*a*** formed in advance.
(32) In the aperture stop plate **24**, various systems can be adopted as the switching mechanism **38** for switching the aperture stops **24*a*** to be arranged in the optical path of the illumination system ILS of the position detection system **16**. As an example, FIG. **3** shows the switching mechanism **38** using a guide rail system that selects a desired aperture stop **24*a*** to be arranged in the optical path of the illumination system ILS by sliding the aperture stop plate **24** in which the plurality of aperture stops **24*a*** are linearly arrayed. After the desired aperture stop **24*a*** is selected, the switching mechanism **38** can also finely adjust the position of the aperture stop **24*a*** in the Z direction by slightly sliding the aperture stop plate **24**. An operation like this can be implemented when the switching mechanism **38** has, for example, a linear/pulse motor.
(33) The direction in which the switching mechanism **38** operates the aperture stop plate **24** need

not be one-dimensional. For example, the switching mechanism **38** can two-dimensionally operate the aperture stop plate **24** in a plane (Z-X plane) perpendicular to the optical axis of the illumination system ILS. Position adjustment (pupil adjustment) in the Z-X plane is performed on the aperture stop **24***a* arranged in the optical path of the illumination system ILS, such that the imaging performance of the whole position detection system **16** improves by combining the aperture stop **24***a* and the aperture stop **34***a* arranged in the optical path of the imaging system IMS.
(34) The light having passed through the aperture stop plate **24** (the aperture stop **24***a*) is guided to the polarizing beam splitter **27** via the third and fourth illumination optical systems **25** and **26**. Of the light guided to the polarizing beam splitter **27**, S-polarized light perpendicular to the paper surface is reflected by the polarizing beam splitter **27** and passes through the NA stop **28** and the prism **29**. The NA stop **28** is a variable NA stop capable of changing the NA by changing the aperture amount under the control of the control unit **17**. The prism **29** is used to branch the optical path to an optical system (not shown). Light from the prism **29** is converted into circularly polarized light through the λ/4 plate **30**, and illuminates the alignment mark **19** formed on the substrate **3** via the objective lens **31**.
(35) Light (detection light) diffracted by the alignment mark **19** is converted into P-polarized light parallel to the paper surface through the objective lens **31** and the λ/4 plate **30**, and transmitted through the polarizing beam splitter **27** via the prism **29** and the NA stop **28**. The light transmitted through the polarizing beam splitter **27** passes through the relay lens **32**, the first imaging optical system **33**, and the aperture stop plate **34**.
(36) The aperture stop plate **34** includes a plurality of aperture stops **34***a* (second stops) having different numerical apertures (aperture (opening) sizes, NA) and different aperture (opening) shapes. The plurality of aperture stops **34***a* are selectively arranged in the pupil plane of the imaging system IMS. In this embodiment, a detection mode for detecting the alignment mark **19** can be changed by switching the optical paths of the imaging system IMS of the position detection system **16**, more specifically, by switching the aperture stops **34***a* to be arranged in the pupil (pupil plane), under the control of the control unit **17**. The aperture stop plate **34** can also have a configuration capable of adding a new aperture stop, in addition to the plurality of aperture stops **34***a* formed in advance.
(37) In the aperture stop plate **34**, various systems can be adopted as the switching mechanism **39** for switching the aperture stops **34***a* to be arranged in the optical path of the imaging system IMS of the position detection system **16**. As an example, FIG. **3** shows the switching mechanism **39** using a turret system that selects a desired aperture stop **34***a* to be arranged in the optical path of the imaging system IMS by rotating (driving), around a rotating shaft **40**, the aperture stop plate **34** in which the plurality of aperture stops **34***a* are arrayed. Note that the plurality of aperture stops **34***a* are arrayed in the aperture stop plate **34** such that the aperture centers are positions on a single circumference. Note also that after the desired aperture stop **34***a* is selected, the switching mechanism **39** can finely adjust the rotating shaft **40** in a direction in the paper surface (a direction perpendicular to the optical axis of the imaging system). Consequently, position adjustment (pupil adjustment) in the X-Y plane is performed on the aperture stop **34***a* arranged in the optical path of the imaging system IMS, such that the imaging performance of the whole position detection system **16** improves by combining the aperture stop **34***a* and the aperture stop **24***a* arranged in the optical path of the illumination system ILS.
(38) In this embodiment, the switching mechanisms **38** and **39** adopt the guide rail system and the turret system as the systems for switching the aperture stops. However, the present invention is not limited to them. For example, another system may also be adopted if switching (selection) of aperture stops and fine adjustment of the selected aperture stop are possible.
(39) The light having passed through the aperture stop plate **34** (the aperture stop **34***a*) arrives at the photoelectric conversion element **37** (for example, an image sensor such as a CCD image sensor) via the second imaging optical system **35** and the wavelength shift difference adjustment optical

member **36**. The photoelectric conversion element **37** detects the light from the alignment mark **19**. The storage time can be prolonged until the intensity of the light exceeds a predetermined threshold value. The control unit **17** controls the storage time of the photoelectric conversion element **37**.
(40) In the exposure apparatus **100**, the control unit **17** decides the driving amount of the driving mechanism **18** for driving the substrate stage **4**, based on the position information of the alignment mark **19** obtained by the position detection system **16**, and performs exposure (overlay exposure) on the substrate **3**.
(41) A method of calibrating the position detection system **16** according to this embodiment and effects obtained by the calibration method will be explained below. First, practical configurations of the switching mechanism **38** for switching the aperture stops **24***a* to be arranged in the optical path of the illumination system ILS of the position detection system **16** and the switching mechanism **39** for switching the aperture stops **34***a* to be arranged in the optical path of the imaging system IMS of the position detection system **16** will be explained.
(42) FIG. **4** is a view showing a practical configuration of the switching mechanism **38** adopting the guide rail system. Referring to FIG. **4**, the aperture stop plate **24** (a light-shielding plate) is attached to a guide rail **41** via bearings **42**, and drivable in the Z direction by a gear **43**. In the aperture stop plate **24**, the plurality of aperture stops **24***a* different in NA and shape are linearly arrayed. In the aperture stop plate **24**, positions (ranges) in which the aperture stops **24***a* should exist are indicated by circles in which positions A to F are written. For example, the aperture stop **24***a* to be arranged in the optical path of the illumination system ILS of the position detection system **16** exists in the position of the circle in which position D is written.
(43) FIG. **5** is a view showing a practical configuration of the switching mechanism **39** adopting the turret system. Referring to FIG. **5**, the aperture stop plate **34** (a light-shielding plate) can be rotated (driven) around the rotating shaft **40** by a gear **40***a* integrated with the rotating shaft **40**. In the aperture stop plate **34**, the plurality of aperture stops **34***a* different in NA and shape are arranged such that their aperture centers are positioned on a single circumference. In the aperture stop plate **34**, positions (ranges) in which the aperture stops **34***a* should exist are indicated by circles in which positions a to g are written. For example, the aperture stop **34***a* to be arranged in the optical path of the imaging system IMS of the position detection system **16** exists in the position of a circle in which position e is written.
(44) Note that the number of the aperture stops **24***a* arrayed in the aperture stop plate **24** and that of the aperture stops **34***a* arranged in the aperture stop plate **34** can be either the same or different. Note also that the shape (aperture shape) of the aperture stops **24***a* and **34***a* need not be circular, and the transmittance of the aperture need not be 100%.
(45) As shown in FIGS. **4** and **5**, a plurality of combinations of the aperture stop **24***a* to be arranged in the optical path of the illumination system ILS of the position detection system **16** and the aperture stop **34***a* to be arranged in the optical path of the imaging system IMS of the position detection system **16** generally exist. However, regardless of a combination of the aperture stops **24***a* and **34***a*, it is necessary to adjust the positions of the aperture stops **24***a* and **34***a* in an optically desirable state, in order to maintain the good imaging state (imaging performance) of the position detection system **16**. This adjustment (calibration) of the position detection system **16** uses two evaluation indices, that is, a lateral shift amount when the alignment mark **19** formed on the substrate **3** is defocused, and the asymmetry of an image of the alignment mark **19** (the degree of asymmetrical distortion of the image).
(46) FIG. **6** is a view showing the state of the position detection system **16** before the aperture stop **24***a* to be arranged in the optical path of the illumination system ILS is adjusted. In FIG. **6**, a circle indicated by the thick line shows a state before the aperture stop **34***a* to be arranged in the optical path of the imaging system IMS is adjusted. Referring to FIG. **6**, before the aperture stop **24***a* is adjusted, the optical axis of light (illumination light) for illuminating the substrate **3** shifts from a direction perpendicular to the substrate **3**, so the direction of 0th-order diffracted light reflected by

the substrate **3** shifts from the center of the aperture stop **34***a* before adjustment. As described previously, the aperture stop **24***a* is positioned in the optical Fourier transform plane with respect to the plane where the substrate **3** is positioned, so adjusting the aperture stop **24***a* is equivalent to adjusting an effective light source, that is, adjusting the angle of light to be incident on the substrate **3** (the incidence angle of the illumination light). Accordingly, it is possible, by driving (adjusting) the aperture stop **24***a* in the state shown in FIG. **6**, to correct (remedy) a component in the driving direction of the optical axis of the inclined illumination light. If this component is left uncorrected, it is obvious that an image of the alignment mark **19** on the substrate laterally shifts when the alignment mark **19** is detected in a different focusing position, resulting in an error in position detection of the alignment mark **19**.

(47) In principle, the 0th-order diffracted light reflected by the substrate **3** can be adjusted to the pupil center on the detection side by driving (adjusting) the aperture stop **34***a* indicated by the thick line in FIG. **6**. Since the aperture stop **34***a* is positioned in the optical Fourier transform plane with respect to (the photoelectric conversion plane of) the photoelectric conversion element **37**, adjusting the aperture stop **34***a* is equivalent to selecting the order of the diffracted light from the substrate **3**, which is detected by the photoelectric conversion element **37**.

(48) Generally, diffracted light from a symmetrical detection target such as the alignment mark **19** is generated by positive-negative symmetry around the 0th-order diffracted light. As described above, therefore, the position detection system **16** can implement a good imaging state having little distortion by adjusting the 0th-order diffracted light to the pupil center on the detection side. If the illumination light is inclined, however, asymmetrical distortion occurs in an image of the alignment mark **19**, and this affects the position detection accuracy of the alignment mark **19**. This is mainly caused by two factors: the alignment mark **19** actually has a three-dimensional thickness; and the reflected light from the substrate **3** passes a region which is away from the optical axis of the imaging system IMS and in which the aberration correction state is low.

(49) It is, therefore, basically necessary to match the aperture stops **24***a* and **34***a* while taking the two evaluation indices, that is, the lateral shift amount when the alignment mark **19** formed on the substrate **3** is defocused, and the asymmetry of an image of the alignment mark **19**. Basically (that is, in an ideal state in which the position detection system **16** has no aberration), “to match the aperture stops **24***a* and **34***a*” means matching their centers. If, however, the position detection system **16** has a large comatic aberration, the aperture stops **24***a* and **34***a* are sometimes slightly shifted from each other so as to block the diffracted light passing the region on the pupil plane where the aberration correction state is low, even though the symmetry of the diffracted light to be taken into the imaging system IMS is damaged.

(50) When introducing and operating the exposure apparatus **100**, a step (adjustment/confirmation step) of adjusting the state of the exposure apparatus **100** and confirming that the state is good is normally provided when the exposure apparatus **100** is shipped and installed. In this adjustment/confirmation step, the initial positions of the aperture stops **24***a* and **34***a* are decided while taking the two evaluation indices, that is, the lateral shift amount when the alignment mark **19** formed on the substrate **3** is defocused, and the asymmetry of an image of the alignment mark **19**, as described previously. For example, the initial positions of the aperture stops **24***a* and **34***a* are decided (adjusted) so that the lateral shift amount and asymmetry of an image of the alignment mark **19** formed on a photoelectric conversion element when the alignment mark **19** is defocused fall within allowable ranges. In this embodiment, data (initial data) to be explained below is obtained for each of the aperture stops **24***a* and **34***a*, so that the data can be used when actually introducing and operating the exposure apparatus **100** having the position detection system **16**.

(51) If (the center of) each of the aperture stops **24***a* and **34***a* shifts from the initial position decided in the adjustment/confirmation step, there are generally four degrees of freedom about driving (the driving amounts) of the aperture stops **24***a* and **34***a* in order to correct the shifts from the initial positions. More specifically, two linearly independent directions in a plane perpendicular to the

optical axis exist for each of the aperture stops **24***a* and **34***a*. Although these two directions need not be perpendicular to each other, they will be represented by using suffixes x and y. Note that when the guide rail system as shown in FIG. **4** is adopted as the switching mechanism for switching aperture stops in one of the illumination system ILS and the imaging system IMS of the position detection system **16**, the number of degrees of freedom about driving of the aperture stops **24***a* and **34***a* is less than four.
(52) In a case in which the aperture stops **24***a* and **34***a* shift from the initial positions when calibrating the position detection system **16** while the exposure apparatus **100** is in operation, the driving amounts (correction driving amounts) of the aperture stops **24***a* and **34***a* are unknowns to be obtained to correct the shifts. Equation 1 below shows them as four-vector.

(53) $$\begin{pmatrix} PILx \\ PDTx \\ PILy \\ PDTy \end{pmatrix} \quad (1)$$

(54) In equation 1, PILx is a correction driving amount in the x direction of the aperture stop **24***a* arranged in the optical path of the illumination system ILS, and PILy is a correction driving amount in the y direction of the aperture stop **24***a* arranged in the optical path of the illumination system ILS. PDLx is a correction driving amount in the x direction of the aperture stop **34***a* arranged in the optical path of the imaging system IMS, and PDLy is a correction driving amount in the y direction of the aperture stop **34***a* arranged in the optical path of the imaging system IMS.
(55) While the driving mechanism **18** is driving the substrate stage **4** in the Z direction, the position detection system **16** detects the alignment mark **19** for adjustment (calibration). The alignment mark **19** for adjustment need not be formed on the substrate **3**, and need only be formed on the substrate stage **4** and driven in the Z direction together with the substrate stage **4**. For example, the alignment mark **19** for adjustment can also be formed on the stage reference plate **11**. When the alignment mark **19** for adjustment is formed on the stage reference plate **11**, the calibration method of this embodiment can be performed in a self-contained manner by using only the constituent elements of the exposure apparatus **100**.
(56) If the optical axis of the illumination system ILS of the position detection system **16** shifts from a vertical direction with respect to the plane on which the alignment mark **19** exists, an image of the alignment mark **19** to be detected by the photoelectric conversion element **37** shifts in the horizontal direction as the substrate stage **4** is driven in the Z direction. Therefore, from the shift directions (x and y directions) and the shift amount of the image of the alignment mark **19** when the substrate stage **4** is driven in the Z direction by a micro unit amount **6**Z, the degree of a shift of the optical axis of light (illumination light) for illuminating the substrate **3** from the vertical direction can be evaluated. In the following explanation, the degree of the shift of the illumination light from the vertical direction with respect to the substrate **3** (that is, the shift amount of the image of the alignment mark **19** on the photoelectric conversion element with respect to the defocus amount of the alignment mark **19**) will be called the defocus characteristic as an evaluation index.
(57) The amount (shift amount) by which (the center of) each of the aperture stops **24***a* and **34***a* shifts from the initial position decided in the adjustment/confirmation step is normally so small that the defocus characteristic appearing as a result can be regarded as proportional to the shift amount. Accordingly, four independent linear relationships need only be used to decide the four correction driving amounts represented by equation 1.
(58) In this embodiment, to obtain the four independent linear relationships (equations) as described above, two different combinations are selected from a plurality of combinations of the aperture stops **24***a* and **34***a*. More specifically, two different combinations of the aperture stops **24***a* and **34***a* are selected by selecting one aperture stop from each of the plurality of aperture stops **24***a* shown in FIG. **4** and the plurality of aperture stops **34***a* shown in FIG. **5**. In this embodiment, these

two combinations will be referred to as mode 1 (the first combination) and mode 2 (the second combination). As the combination of the aperture stops **24***a* and **34***a* to be selected as mode 1 or mode 2 can also be selected from the existing combinations for implementing illumination modes already set in the exposure apparatus **100**, in accordance with a purpose to be explained below. Note that as will be described later, it is also possible to exclusively set an aperture stop combination having a structure optimized for the purpose of calibrating the position detection system **16**.
(59) When the two combinations for implementing mode 1 and mode 2 are selected as described above, the fluctuation amount of the defocus characteristic can be represented by four-vector as indicated by equation 2 below:

(60) $$\begin{pmatrix} 1x \\ \Delta 2x \\ 1y \\ \Delta 2y \end{pmatrix} \quad (2)$$

(61) In equation 2, Δ1x indicates the fluctuation amount of the defocus characteristic in the x direction in mode 1, and Δ2x indicates the fluctuation amount of the defocus characteristic in the x direction in mode 2. Also, Δ1y indicates the fluctuation amount of the defocus characteristic in the y direction in mode 1, and Δ2y indicates the fluctuation amount of the defocus characteristic in the y direction in mode 2. FIG. **7** shows the concept indicating that these fluctuation amounts of the defocus characteristic are proportional to the driving amount of the substrate stage **4** in the Z direction. FIG. **7** is a view showing the relationship between the driving amount of the substrate stage **4** in the Z direction and the fluctuation amount of the defocus characteristic. In FIG. **7**, the abscissa indicates the driving amount of the substrate stage **4** in the Z direction, and the ordinate indicates (the fluctuation amount of) the defocus characteristic.
(62) In the adjustment/confirmation step of this embodiment, the values (D01x, D02x, D01y, and D02y) of the defocus characteristics in the x and y directions in modes 1 and 2 when the initial adjustment is complete are measured and obtained as initial data (reference defocus characteristics). In addition, fluctuation amount vectors of the defocus characteristic when fluctuating the aperture stops **24***a* and **34***a* by a unit amount in the pupil planes of the illumination system ILS and the imaging system IMS are obtained as initial data by executing procedures (1), (2), (3), and (4) to be explained below. Procedure (1): When only the aperture stop **24***a* arranged in the optical path of the illumination system ILS of the position detection system **16** is fluctuated by a unit amount in the x direction, fluctuation vectors (Δ1x, Δ2x, Δ1y, and Δ2y) of the defocus characteristic are obtained as (A, B, C, and D). Procedure (2): When only the aperture stop **34***a* arranged in the optical path of the imaging system IMS of the position detection system **16** is fluctuated by a unit amount in the x direction, fluctuation vectors (Δ1x, Δ2x, Δ1y, and Δ2y) of the defocus characteristic are obtained as (E, F, G, and H). Procedure (3): When only the aperture stop **24***a* arranged in the optical path of the illumination system ILS of the position detection system **16** is fluctuated by a unit amount in the y direction, fluctuation vectors (Δ1x, Δ2x, Δ1y, and Δ2y) of the defocus characteristic are obtained as (I, J, K, and L). Procedure (4): When only the aperture stop **34***a* arranged in the optical path of the imaging system IMS of the position detection system **16** is fluctuated by a unit amount in the y direction, fluctuation vectors (Δ1x, Δ2x, My, and Δ2y) of the defocus characteristic are obtained as (M, N, O, and P).
(63) When calibrating the position detection system **16** (when adjusting the positions of the aperture stops **24***a* and **34***a*) in a case where the operation of the exposure apparatus **100** is started and the aperture stops **24***a* and **34***a* shift from the initial positions decided in the adjustment/confirmation step, a procedure to be explained below is executed.
(64) First, in a state in which the aperture stops **24***a* and **34***a* exist in positions (first positions) shifted from the initial positions, values (D1x, D2x, D1y, and D2y) of the defocus characteristics in

the x and y directions in modes 1 and 2 are measured and obtained. Consequently, the correction driving amounts (PILx, LDTx, PILy, and DTy) of the aperture stops **24***a* and **34***a* as the unknowns in equation 1 can be calculated from simultaneous equations with four unknowns indicated by equation 3 below:

(65) $$\begin{pmatrix} D1x - D01x \\ D2x - D02x \\ D1y - D01y \\ D2y - D02y \end{pmatrix} = \begin{pmatrix} A & E & K & M \\ B & F & L & N \\ C & G & K & O \\ D & H & L & P \end{pmatrix} \begin{pmatrix} P_{\mathrm{ILx}} \\ P_{\mathrm{DTx}} \\ P_{\mathrm{ILy}} \\ P_{\mathrm{DTy}} \end{pmatrix} \quad (3)$$

(66) The solution of the simultaneous equations with four unknowns indicated by equation 3 is simple. Therefore, the equations can be solved by an arithmetic unit in the exposure apparatus **100**, for example, the control unit **17**, and can also be solved by an external apparatus (arithmetic device) different from the exposure apparatus **100**.
(67) The switching mechanisms **38** and **39** are controlled (driven) based on the correction driving amounts calculated from the simultaneous equations with four unknowns indicated by equation 3, thereby adjusting the positions of the aperture stops **24***a* and **34***a* (returning them to the initial positions), and achieving the calibration of the position detection system **16**.
(68) In this embodiment, the initial positions decided in the adjustment/confirmation step are set as the reference positions of the aperture stops **24***a* and **34***a*, and, if the aperture stops **24***a* and **34***a* shift from the initial positions, the aperture stops **24***a* and **34***a* are simply returned to the initial positions. However, the reference positions as the positions to which the aperture stops **24***a* and **34***a* are returned are not limited to the initial positions, and can be set at arbitrary positions. Also, if there is a purpose of intentionally changing the characteristics of the position detection system **16** in order to cope with the specialty of an individual process substrate, the left side of equation 3 can be set as the value (target value) of the defocus characteristic corresponding to an arbitrary target.
(69) In this embodiment, the number of combinations (modes) of the aperture stops **24***a* and **34***a* necessary to calibrate the position detection system **16** is at least two. However, it is also possible to execute the above-described procedures by preparing more than two combinations, and obtain the correction driving amounts of the aperture stops **24***a* and **34***a* by comprehensively taking the results into consideration. For example, when N combinations are prepared, N(N−1)/2 correction driving amounts (PILx, LDTx, PILy, and DTy) are obtained, so the weighted average values of these vectors can be adopted.
(70) Also, alignment light (non-exposure light) used in the exposure apparatus **100** generally contains a plurality of wavelength bands, for example, at least two wavelength bands of 450 nm or more, in order to cope with various processes. In a case like this, weighted averaging by the wavelength can be performed.
(71) Next, a selection (combination) of the aperture stops **24***a* and **34***a* suitable when actually performing the calibration method of the position detection system **16** according to this embodiment will be explained. As is apparent from equation 3, this embodiment calculates the correction driving amounts of the aperture stops **24***a* and **34***a* from the linear equations (the simultaneous equations with four unknowns). Accordingly, if the value of the determinant of a coefficient matrix indicated by equation 4 below is close to zero, the accuracy of the correction driving amounts (PILx, LDTx, PILy, and DTy) of the aperture stops **24***a* and **34***a* calculated from equation 3 decreases. Therefore, the value of the determinant of the coefficient matrix indicated by equation 4 must be set at a value away from zero.

(72) .Math. $$\begin{matrix} A & E & K & M \\ B & F & L & N \\ C & G & K & O \\ D & H & L & P \end{matrix}$$ .Math. (4)

(73) An alignment mark **19**A as shown in FIG. **8** can be used as the alignment mark **19** that can be used in the exposure apparatus **100** (the position detection system **16**) and can also be simplified. FIG. **8** is a view showing the alignment mark **19**A including a structure that generates diffracted light in only one of two directions perpendicular to each other in a plane (X-Y plane) perpendicular to the optical axis of the position detection system **16**. The alignment mark **19**A generates diffracted light in only the Y direction of the coordinate axes shown in FIG. **8**, due to the symmetry of the mark. Accordingly, even when the aperture stops **24***a* and **34***a* are driven in the X direction in the coordinate axes shown in FIG. **8**, the defocus characteristic in the X direction probably hardly fluctuates. By using the alignment mark **19**A with which the direction in which no diffracted light is generated and the direction in which diffracted light is generated respectively match the X and Y directions in the position detection system **16** as described above, equation 4 is accurately approximated to a block diagonalization matrix as indicated by equation 5 below:

(74) $$\begin{pmatrix} A & E & 0 & 0 \\ B & F & 0 & 0 \\ 0 & 0 & K & O \\ 0 & 0 & L & P \end{pmatrix} \quad (5)$$

(75) When the X and Y directions are separated, the determinant indicated by equation 5 is (AF−BE).Math.(KP−LO), that is, the product of the determinants of submatrices existing in the diagonal positions. Therefore, modes 1 and 2 (the combination of the aperture stops **24***a* and **34***a*) need only be so selected as to set the values of AF−BE and KP−LO apart from zero at the same time.
(76) It is possible to prepare two combinations of the aperture stops **24***a* and **34***a* as described above. Assume, for example, that the aperture stops **24***a* and **34***a* have apertures (openings) that are perfect circles (almost perfect circles) having a common center (that is, the positions of the centers overlap each other), and having different radii. In this case, it is only necessary to combine (select) the aperture stops **24***a* and **34***a* so that the aperture stops **24***a* are largely different in only the radius of the aperture (aperture (opening) radius) in modes 1 and 2, and the aperture stops **34***a* have a common (the same) aperture radius in modes 1 and 2. More specifically, the aperture (opening) ratio of the aperture stop **24***a* included in mode 1 to the aperture stop **24***a* included in mode 2 is set to be twice or more. In addition, the aperture stops **34***a* included in modes 1 and 2 have an aperture radius larger than those of the aperture stops **24***a* included in modes 1 and 2. A combination like this is optically already known. For example, the mode including the aperture stop **24***a* having a large aperture radius is a mode giving importance to the resolving power, and the mode including the aperture stop **34***a* having a small aperture radius is a mode giving importance to the contrast.
(77) “The aperture radius of the aperture stop **24***a* is small” means that the radius of light passing through the aperture stop **34***a* is small, so the light passes inside the aperture stop **34***a*. In the mode giving importance to the contrast, therefore, the influence of the shift of the aperture stop **34***a* on the defocus characteristic is extremely small, compared to the mode giving importance to the resolving power. Since this means that the values of F and P are close to zero in equation 5, (AF−BE).Math.(KP−LO)≠BE.Math.LO holds, so it is possible to avoid the value of the whole determinant from becoming close to zero.
(78) Table 1 below shows practical numerical value examples of the aperture stops **24***a* and **34***a* in modes 1 and 2 when the aperture stops **24***a* and **34***a* have perfectly circular apertures having a common center and different radii as described above.
(79) TABLE-US-00001 TABLE 1 Aperture Aperture Stop 24a Stop 34a (NA) (NA) Mode 1 0.35 0.4 Mode 2 0.15 0.4
(80) The process (calibration method) of calibrating the position detection system **16** will be explained below with reference to FIG. **9**. The control unit **17** performs this process by comprehensively controlling the units of the exposure apparatus **100** including the position

detection system **16**. In other words, the control unit **17** also functions as a processing unit for performing the process of calibrating the position detection system **16**. Instead of the control unit **17**, however, an external apparatus (information processing apparatus) can also perform the process of calibrating the position detection system **16** by comprehensively controlling the units of the exposure apparatus **100** including the position detection system **16**.

(81) In this embodiment, the exposure apparatus **100** is so configured as to be able to calibrate the position detection system **16** at any arbitrary timing while the exposure apparatus **100** is in operation. However, the aperture stops **24***a* and **34***a* actually shift from the initial positions decided in the adjustment/confirmation step in a case where the supply of power to the exposure apparatus **100** is shut down and resumed after that (at the time of power refeeding). Also, the aperture stops **24***a* and **34***a* may shift from the initial positions decided in the adjustment/confirmation step when the exposure apparatus **100** is reset (at the time of resetting) for an operation of returning from some sort of trouble. Accordingly, the effectiveness of the operation of the exposure apparatus **100** can be increased by calibrating the position detection system **16** at the time of refeeding of power or resetting as described above. Note that even at the time of refeeding of power or resetting, the user can determine not to calibrate the position detection system **16**.

(82) Referring to FIG. **9**, in S**101**, the current defocus characteristics (D1x and D1y) (after the aperture stops **24***a* and **34***a* have shifted from the initial positions) are obtained (measured) in preselected mode 1. More specifically, the switching mechanisms **38** and **39** are driven to arrange the aperture stop **24***a* selected in mode 1 in the pupil plane of the illumination system ILS, and arrange the aperture stop **34***a* selected in mode 1 in the pupil plane of the imaging system IMS. In addition, the substrate stage **4** is driven such that the alignment mark **19** is positioned within the visual field of the position detection system **16**. Then, while the substrate stage **4** is driven in the Z direction, the current defocus characteristics in mode 1 are obtained by detecting the alignment mark **19** by the position detection system **16**.

(83) In S**102**, the current defocus characteristics (D2x and D2y) (after the aperture stops **24***a* and **34***a* have shifted from the initial positions) are obtained (measured) in preselected mode 2. More specifically, the switching mechanisms **38** and **39** are driven to arrange the aperture stop **24***a* selected in mode 2 in the pupil plane of the illumination system ILS, and arrange the aperture stop **34***a* selected in mode 2 in the pupil plane of the imaging system IMS. In addition, the substrate stage **4** is driven such that the alignment mark **19** is positioned within the visual field of the position detection system **16**. Then, while the substrate stage **4** is driven in the Z direction, the current defocus characteristics in mode 2 are obtained by detecting the alignment mark **19** by the position detection system **16**.

(84) In this embodiment, in S**101** and S**102** (in the first step), the current defocus characteristics (first defocus characteristics) are obtained in each of modes 1 and 2.

(85) In S**103**, differences between the defocus characteristics (D01x, D02x, D01y, and D02y) in modes 1 and 2 obtained in advance in the adjustment/confirmation step, and the current defocus characteristics (D1x, D2x, D1y, and D2y) in modes 1 and 2 obtained in S**101** and S**102**, are obtained.

(86) In S**104**, the driving amounts of the aperture stops **24***a* and **34***a* necessary to correct the shifts from the initial positions of the aperture stops **24***a* and **34***a*, that is, the correction driving amounts (PILx, PILy, PDLx, and PDLy) are decided. As described previously, the correction driving amounts of the aperture stops **24***a* and **34***a* can be calculated from the simultaneous equations with four unknowns indicated by equation 3 based on the differences obtained in S**103**. Note that various kinds of data including initial data required to calculate the correction driving amounts of the aperture stops **24***a* and **34***a*, that is, required to solve the simultaneous equations with four unknowns indicated by equation 3 can be read out from a storage area formed in the exposure apparatus **100**, and can also be read out from an external storage area.

(87) In S**105**, the aperture stop **24***a* (the switching mechanism **38**) and the aperture stop **34***a* (the

switching mechanism **39**) are driven in accordance with the correction driving amounts (PILx, PILy, PDLx, and PDLy) decided in S**104**, thereby adjusting the positions of the aperture stops **24***a* and **34***a*.
(88) In this embodiment, in S**103**, S**104**, and S**105** (in the second step), position adjustment (pupil adjustment) of each of the aperture stops **24***a* and **34***a* is performed based on the reference defocus characteristics and the current defocus characteristics in modes 1 and 2.
(89) As described above, according to the calibration method of the position detection system **16** in this embodiment, the position detection system **16** can be calculated on the user site at an arbitrary timing in a self-contained manner without requiring any special tool in addition to the exposure apparatus **100**. Also, since an encoder for detecting shifts (pupil position shifts) of the aperture stops **24***a* and **34***a* is unnecessary, it is possible to decrease the cost and remove a heat source at the same time, and improve the position detection accuracy. According to this embodiment, therefore, the exposure apparatus **100** superior in cost and overlay accuracy can be provided.
(90) A method of manufacturing an article according to the embodiment of the present invention is suitable for manufacturing an article such as a flat panel display, a liquid crystal display element, a semiconductor device, a MEMS or the like. This method of manufacturing includes a step of exposing a substrate coated with a photosensitive agent by using the above-described exposure apparatus **100** and a step of developing the exposed photosensitive agent. An etching step and an ion implantation step are performed on the substrate by using the pattern of the developed photosensitive agent as a mask, thereby forming a circuit pattern on the substrate. By repeating the steps such as these exposure, development, and etching steps, a circuit pattern formed from a plurality of layers is formed on the substrate. In a subsequent step, dicing (processing) is performed on the substrate on which the circuit pattern has been formed, and mounting, bonding, and inspection steps of a chip are performed. The method of manufacturing can further include other known steps (oxidation, deposition, vapor deposition, doping, planarization, resist removal, and the like). The method of manufacturing the article according to this embodiment is superior to the conventional method in at least one of the performance, quality, productivity, and production cost of the article.
(91) The present invention can also be implemented by a process of supplying a program that implements one or more functions of the above-described embodiment to a system or an apparatus via a network or a storage medium, and causing one or more processor in a computer of the system or apparatus to read out and execute the program. Furthermore, the present invention can also be implemented by a circuit (for example, an ASIC) that implements one or more functions.
(92) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(93) This application claims the benefit of Japanese Patent application No. 2022-070268 filed on Apr. 21, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. A calibration method of a detection system including an illumination system configured to illuminate a detection target, and an imaging system configured to form an image of light from the detection target on a photoelectric conversion element, wherein the illumination system includes a plurality of first apertures selectively arranged in a pupil plane of the illumination system and having different openings, the imaging system includes a plurality of second apertures selectively arranged in a pupil plane of the imaging system and having different openings, and the calibration method comprises: obtaining, for each of at least two combinations of first apertures and second apertures, each of which is formed by selecting one first aperture and one second aperture from the

plurality of first apertures and the plurality of second apertures a first defocus characteristic indicating a shift amount of the image on the photoelectric conversion element with respect to a defocus amount of the detection target in a state in which each of the first aperture and the second aperture is positioned in a first position shifted from a reference position, and performing position adjustment on each of the first aperture and the second aperture such that each of the first aperture and the second aperture is positioned in the reference position, based on a reference defocus characteristic indicating a shift amount of the image on the photoelectric conversion element with respect to a defocus amount of the detection target in a state in which each of the first aperture and the second aperture is positioned in the reference position, and on the first defocus characteristic obtained in the obtaining, wherein in the performing the position adjustment, a driving amount of each of the first aperture and the second aperture required for the position adjustment is obtained from a difference between the first defocus characteristic and the reference defocus characteristic, and from a fluctuation amount of the reference defocus characteristic when the first aperture and the second aperture are fluctuated by a unit amount in the pupil plane of the illumination system and the pupil plane of the imaging system.
2. The method according to claim 1, wherein the reference defocus characteristic and the fluctuation amount of the reference defocus characteristic are obtained in advance before the obtaining is performed.
3. The method according to claim 1, wherein the first aperture arranged in the pupil plane of the illumination system and the second aperture arranged in the pupil plane of the imaging system have perfectly circular openings having central positions overlapping each other and different radii.
4. The method according to claim 1, wherein the at least two combinations include a first combination and a second combination, and an opening ratio of a first aperture, of the plurality of first apertures, included in the first combination to a first aperture, of the plurality of first apertures, included in the second combination is not less than twice.
5. The method according to claim 1, wherein the at least two combinations include a first combination and a second combination, a second aperture, of the plurality of second apertures, included in the first combination is also included as the second aperture, of the plurality of second apertures, included in the second combination, and the second aperture included in the first combination and in the second combination has an opening radius larger than an opening radius of a first aperture, of the plurality of first apertures, included in the first combination and a first aperture, of the plurality of first apertures, included in the second combination.
6. The method according to claim 1, wherein the light is light including at least two types of wavelength bands of not less than 450 nm.
7. The method according to claim 1, wherein the detection target includes a structure configured to generate diffracted light in only one of two directions perpendicular to each other in a plane perpendicular to an optical axis of the detection system.
8. The method according to claim 1, wherein the reference position includes an initial position in which a position of each of the first aperture and the second aperture is adjusted such that a lateral shift amount and asymmetry of the image formed on the photoelectric conversion element when the detection target is defocused fall within allowable ranges.
9. A non-transitory computer-readable storage medium storing a program for causing a computer to execute the calibration method defined in claim 1.
10. A detection system for detecting a position of a detection target, comprising: an illumination system configured to illuminate the detection target; an imaging system configured to form an image of light from the detection target on a photoelectric conversion element; and a processing unit configured to perform a process of calibrating the detection system, wherein the illumination system includes a plurality of first apertures selectively arranged in a pupil plane of the illumination system and having different openings, the imaging system includes a plurality of second apertures selectively arranged in a pupil plane of the imaging system and having different

openings, and the processing unit obtains, for each of at least two combinations of first apertures and second apertures, each of which is formed by selecting one first aperture and one second aperture from the plurality of first apertures and the plurality of second apertures, a first defocus characteristic indicating a shift amount of the image on the photoelectric conversion element with respect to a defocus amount of the detection target in a state in which each of the first aperture and the second aperture is positioned in a first position shifted from a reference position, and performs position adjustment on each of the first aperture and the second aperture such that each of the first aperture and the second aperture is positioned in the reference position, based on a reference defocus characteristic indicating a shift amount of the image on the photoelectric conversion element with respect to a defocus amount of the detection target in a state in which each of the first aperture and the second aperture is positioned in the reference position, and on the first defocus characteristic, wherein in the position adjustment performed by the processing unit, a driving amount of each of the first aperture and the second aperture required for the position adjustment is obtained from a difference between the first defocus characteristic and the reference defocus characteristic, and from a fluctuation amount of the reference defocus characteristic when the first aperture and the second aperture are fluctuated by a unit amount in the pupil plane of the illumination system and the pupil plane of the imaging system.
11. An exposure apparatus for exposing a substrate, comprising: a detection system defined in claim 10 and configured to detect a mark formed on the substrate as a detection target; and a control unit configured to adjust a position of the substrate based on a position of the mark detected by the detection system.
12. An article manufacturing method comprising: exposing a substrate using an exposure apparatus defined in claim 11; developing the exposed substrate; and manufacturing an article from the developed substrate.